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▽ Guide: The Missing Pieces in Recursive Self-Improvement

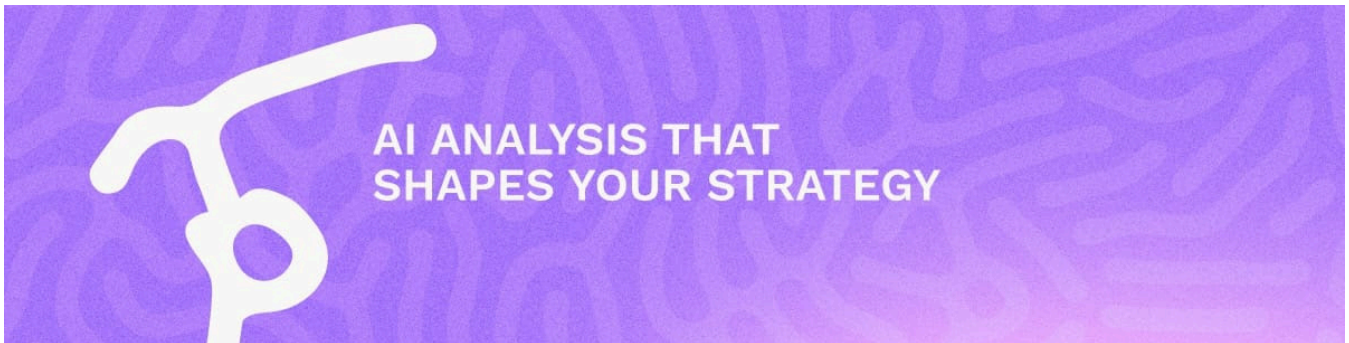
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▽ Guide: The Missing Pieces in Recursive Self-Improvement

AI can improve AI. Why hasn't recursive self-improvement happened yet?



Alyona Vert.



*Housekeeping, folks: AI 101 is becoming ▽ Guide and moving to **Friday**. We're going beyond the basics of AI concepts, models, and techniques, connecting these explanations to our Almanac. The symbol is called nabla, used in mathematics to express a gradient. A small nod to how models learn, as we keep learning ourselves.*

RSI is an abbreviation that has become all too familiar over the past few months. That's how we describe it: **Recursive self-improvement (RSI) is the idea that AI can help improve future AI systems, which can then help build even better ones.**

For years, developers have worked to improve model capabilities. Now, as models become more capable, the temptation is to hand more of that work over to them. The promise is just too compelling: models that could

- learn from experience, much as humans get better at their jobs
- help design and train stronger successors
- adapt to unfamiliar problems without waiting for human updates.

These capabilities could accelerate scientific and engineering progress beyond anything we have seen before. And it's hard not to get excited about AI agents becoming first-class contributors to science.

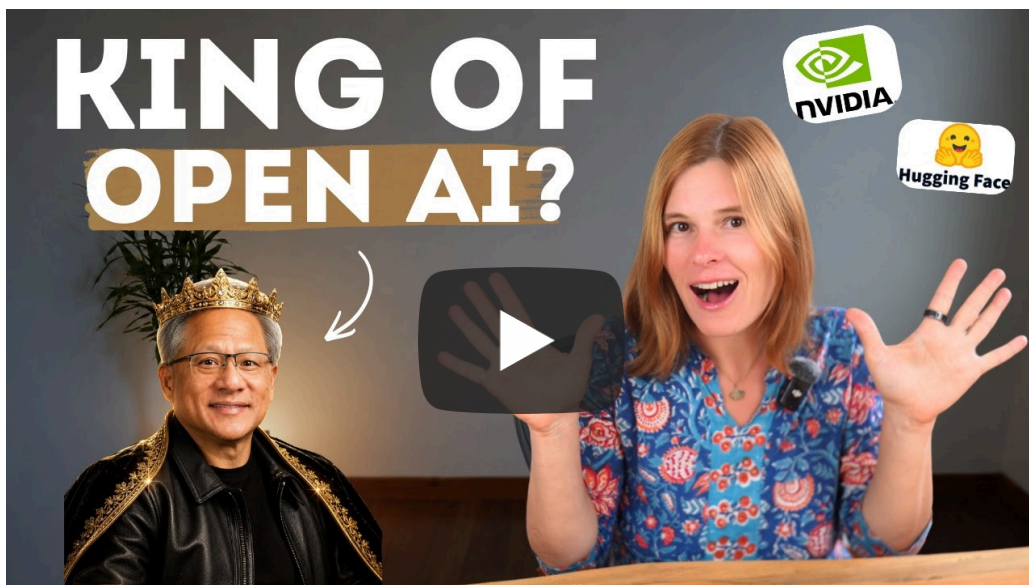
Today, we already see that AI *can* improve AI (we are going to publish an interesting interview about that next week). But why hasn't that turned into sustained recursive self-improvement? A lot depends on the answer, including how quickly we can expect AI to advance.

In this article, we'll look at what's holding RSI back. Even if RSI isn't your thing, these bottlenecks are worth understanding because many of them slow down AI development more broadly. So let's figure out where progress gets a little bit stuck.

In today's episode:

- *What limits Recursive Self-Improvement? Iteration Is Not the Same as Rethinking*
- *Why Iteration Matters for RSI but Is Not Enough*
- *Metaⁿ: A Layered Approach to Improving RSI*
- *Recuris: The Power of Memory in RSI*
- *Conclusion: Everything is a puzzle piece*
- *Sources and further reading*

Not interested in this topic? [In this video](#) we examine how NVIDIA went from nearly dying with NV1 to finding CUDA's market in deep learning, and how that journey led to its reported \$12.9 billion acquisition of Hugging Face. Has NVIDIA figured out how to make open-source AI extraordinarily profitable?



NVIDIA's \$12.9B Plan to Rule Open-Source AI

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What limits Recursive Self-Improvement? Iteration Is Not the Same as Rethinking

We'll start our exploration with a paper that, at first glance, doesn't seem to be about RSI: ***"What is Missing from AI Post-Training AI: An Empirical Analysis"*** by researchers from Tsinghua University and other institutions. It looks at AI agents post-training other AI models. This may sound like a narrow training problem, but it gives us a concrete way to see how far AI can already go in improving AI, and what is still missing from that process.

Researchers separate an agent's ability into two levels:

1. **Execution** – the agent follows an existing training plan. It prepares data, tunes settings, fixes bugs, adjusts rewards, and chooses the best checkpoint.
2. **Strategy** – the agent changes the plan itself. It might switch to another training method, add or remove a training stage, use a different type of data, or redirect its remaining compute.

The researchers counted something as a training experiment only when the agent actually updated the model's parameters. Then they looked at what changed between experiments:

- Small changes within the same plan like tuning hyperparameters were counted as execution changes.
- Changes to the training method were counted as strategy changes.

The results of their experiment unveiled an important insight. **Agents already succeed at training models.** Across 1,338 trajectories, they completed 5,111 training runs and improved average benchmark performance from 10.4% to 23.0%. On HumanEval, for example, average performance increased from 22.0% to 41.4%. **So agents can do the work: prepare data, launch training, diagnose technical problems, repair failed runs and iterate for hours.**

The main limitation appeared at the strategy level: agents usually pick a training strategy at the very beginning and tend to stay there, spending the rest of their time on making small adjustments. Agents tested a different strategy in only about 2% of cases. They don't change the method of improvement even when the results suggest they should try something different.

It is interesting that the preferred approach often depended more on the agent than on the task. While Claude Code mostly chose full-parameter supervised fine-tuning, Codex CLI "preferred" parameter-efficient fine-tuning.

The researchers suggested that there has to be a way to solve this. They tried three fixes but none of them work toward the goal:

- **More experience** like giving agents memory, useful skills, and feedback improves only their execution. Agents achieved better scores: for example, HumanEval performance increased from 22.0% to 62.8% (compared to 41.4% without the fix). However, more experience doesn't make agents change their strategy when needed.

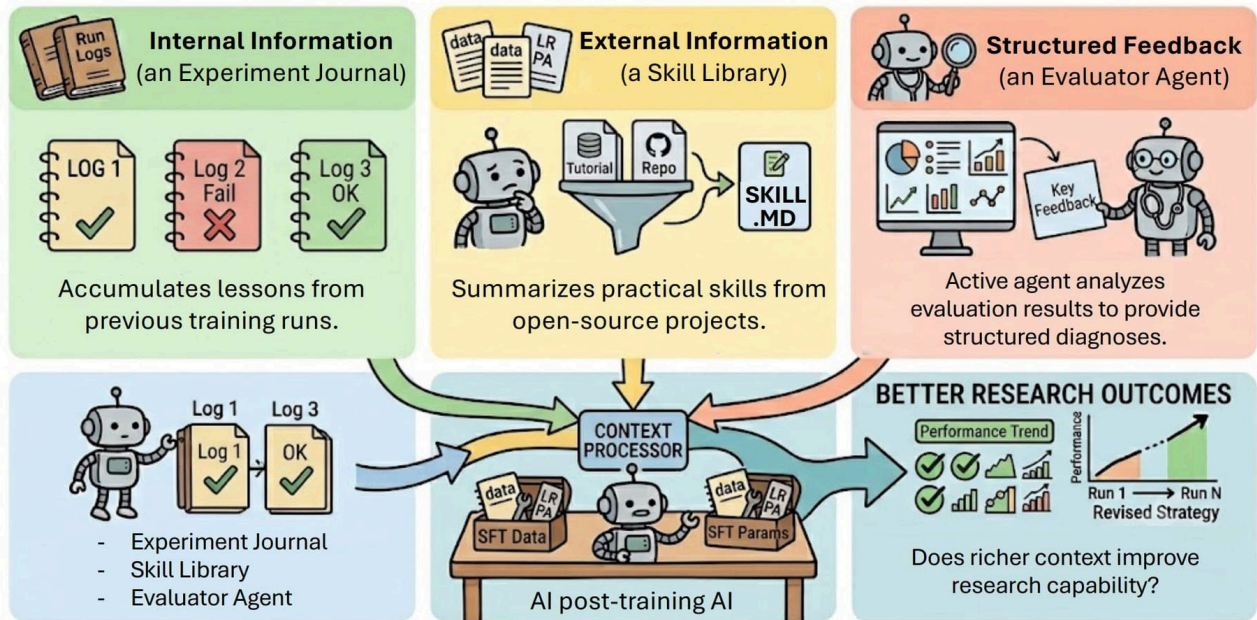


Figure 9: Conceptual overview of the experience-driven agent framework.

Image Credit: "What is Missing from AI Post-Training AI: An Empirical Analysis" paper

- **Human guidance** helps agents choose a different initial approach at the beginning. But once training begins, they again become stuck in small local adjustments.

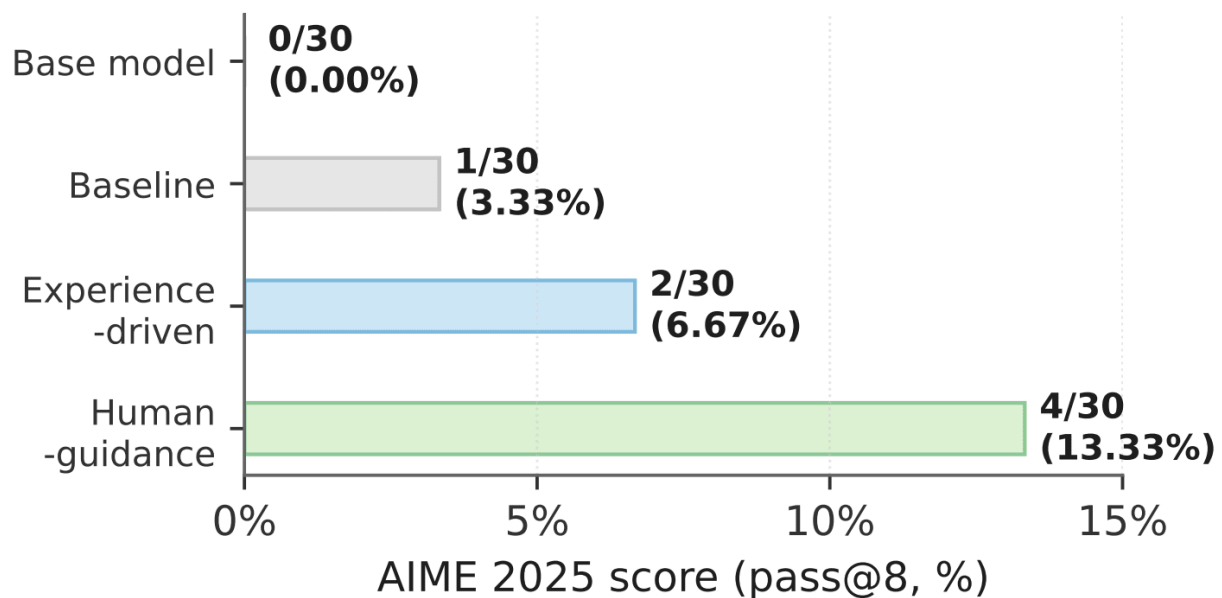


Figure 5: AIME 2025 pass@8 accuracy of final submitted checkpoints for the best run of each setting: base model, autonomous baseline, experience-driven, and human-guided.

Image Credit: “What is Missing from AI Post-Training AI: An Empirical Analysis” paper

- And finally, giving agents 2–8x **more inference compute** helps on easier tasks, but makes almost no difference on the hardest ones.

The problem, then, is not a lack of experience, guidance, or compute. AI can already iterate to improve AI. **But being able to iterate is not the same as being able to rethink.** Current agents rarely can stop and “think”: maybe my whole approach is wrong? They just seem to lack a reliable way to recognize a mistake during the iteration process, so they risk to keep iterating in the wrong direction and get stuck in the cycle of local unnecessary corrections.

So it is bigger than a post-training problem. It is an obstacle for RSI →

Why Iteration Matters for RSI but Is Not Enough

In recursive self-improvement, AI becomes part of the process that improves future AI systems. The key word here is **recursive**: the results of one research cycle are fed back into the next cycle, helping the system choose better options for improvement. To iterate means to repeat automated research loop while learning from each attempt:

Propose an idea → implement it → run an experiment → evaluate the results → **revise the plan** → try again.

This loop becomes recursive only when AI uses what it learns to rethink the strategy behind the next round including the training method, data setup, evaluation process, or model-building pipeline.

Now most agents complete this loop at the execution level. But the broader process of “AI improving AI” remains mostly linear.

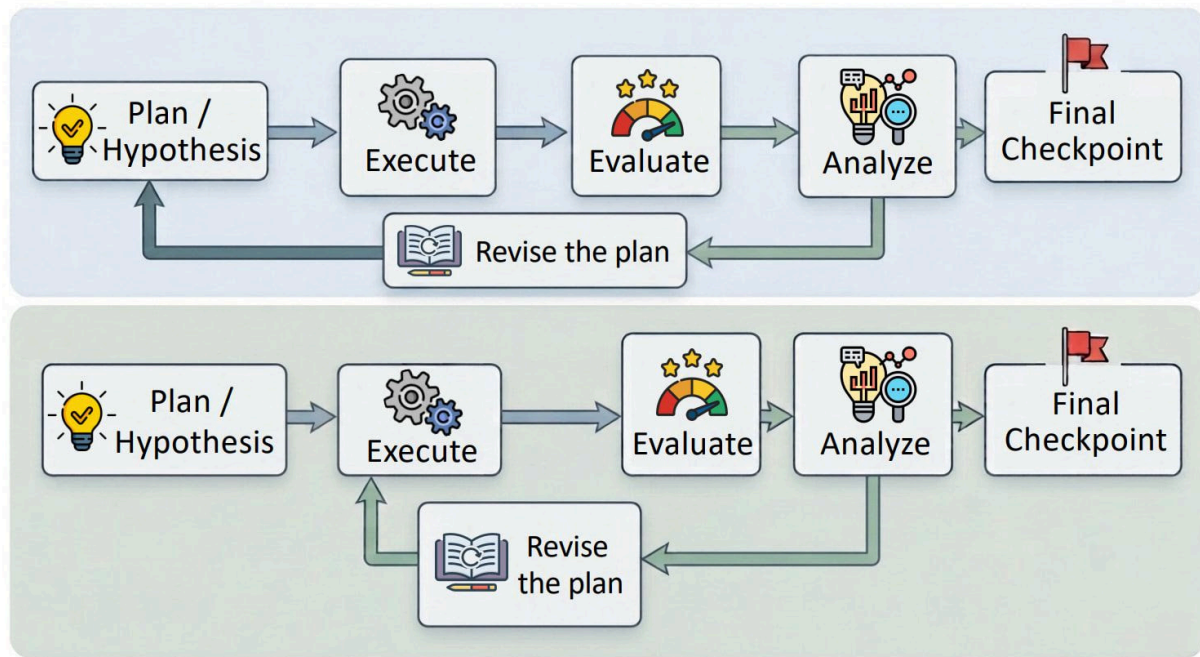


Figure 7: (a) An effective post-training assumes a globally closed loop. (b) Agents operate in local execution loops along a largely linear trajectory.

Image Credit: “What is Missing from AI Post-Training AI: An Empirical Analysis” paper

The “recursive” part is still limited: agents keep working within the original plan. They can try a different approach when a human points them toward it. But **they don’t make that change on their own.**

A bigger model may not solve this. Agents may need to be trained to recognize when an approach isn’t working, with checkpoints where they stop and ask whether to keep going or try something else.

That’s why the researchers pay so much attention to strategy. It determines what the agent tries next. **If the approach is wrong, another round of work may just take it further in the wrong direction.**

Fortunately, there is another fascinating study that proposes a practical solution to improve the process of RSI. →

Metaⁿ : A Layered Approach to Improving RSI

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